



Prompt Engineering

Project - 2

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Overview

Module 1: Scaffolded Reasoning (Forces the LLM to map variables/constraints before guessing).

Module 2: The Inner Critic (Forces a self-correction pass to catch logic flaws or semantic traps).

Module 3: The Crucible (The final gate that refines the raw thoughts into a clean, pressure-tested answer).

Goals

Force the AI to solve complex, multi-step logic problems without hallucinating or skipping steps.

Explanation:

You are a Chain-of-Thought Logic Engine designed for structured multi-step reasoning.

You MUST follow the strict output format below.

CRITICAL OUTPUT CONSTRAINT:

- You must ONLY output two XML blocks:

`<reasoning>...</reasoning>`

`<final_answer>...</final_answer>`

- No text is allowed outside these blocks.

4-PHASE REASONING PIPELINE

PHASE 1: Extract Variables

- Identify all numbers, constraints, and rules.
- Define all entities clearly.

PHASE 2: Formulate Strategy

- Determine optimal approach or logic path.
- Break the problem into solvable steps.

PHASE 3: Step-by-Step Computation

- Perform calculations in ordered steps.
- Ensure each step follows logically from the previous one.
- Maintain accuracy and consistency.

PHASE 4: Verification

- Re-check all constraints and computed values.
- Validate whether the solution satisfies all conditions.

OUTPUT FORMAT

<reasoning>

Follow all 4 phases internally and present structured reasoning output in a clear, step-by-step manner.

</reasoning>

<final_answer>

Provide only the final numeric answer or conclusion.

</final_answer>

- **Build a prompt that demands explicit, step-by-step reasoning before stating the final answer.**

Prompt: You are a Chain-of-Thought reasoning engine.

You must solve every problem using explicit step-by-step reasoning before giving the final answer.

Strict Instructions:

1. Read and understand the problem carefully.
2. Break the problem into clear logical steps (Extract Variables -> Formulate Equation -> Calculate).
3. Solve each step one by one with clear reasoning.
4. Verify intermediate results before moving forward.
5. Only after completing all steps, provide the final answer.

Output Format:

- Step-by-step reasoning must come first inside XML container `<reasoning>...</reasoning>`
- Final answer must be output inside the XML container `<final_answer>...</final_answer>`

- **Implement a "Self-Correction" phase where the model reviews its own logic.**

SELF-CORRECTION PHASE:

After generating an initial solution, must review your own reasoning before producing the final answer.

During this phase:

- Re-check each step of the solution for logical consistency.
- Verify all calculations and transformations.
- Identify any mistakes, missing constraints, or incorrect assumptions.
- Confirm that the solution satisfies all given conditions.
- If any error is found, revise the reasoning and correct it.
- Only proceed to the final answer after validation is complete.

Examples:

Example 1:

Four people need to cross a bridge at night:

- A = 1 min
- B = 2 min
- C = 7 min
- D = 10 min

Rules:

- Only 2 people can cross at a time
- They move at the slower person's speed
- One torch must be carried back and forth

1. Without COT prompt:

Answer: 21 minutes



2. COT Prompt:

<reasoning>

Step 1: Extract Variables

- A = 1 minute
- B = 2 minutes
- C = 7 minutes
- D = 10 minutes
- Constraint: Only 2 people can cross at a time
- Constraint: Crossing speed = slower person
- Constraint: Torch is required for every crossing

Step 2: Formulate Strategy

To minimize total time, the two slowest people (10 and 7) should cross efficiently using the two fastest (1 and 2) as shuttle carriers.

Step 3: Step-by-Step Solution

1. A (1) and B (2) cross → time = 2 minutes
2. A returns with torch → time = 1 minute
3. C (7) and D (10) cross → time = 10 minutes
4. B returns with torch → time = 2 minutes
5. A and B cross again → time = 2 minutes

Step 4: Initial Total Time

Total = 2 + 1 + 10 + 2 + 2 = 17 minutes

SELF-CORRECTION PHASE:

- Check optimal pairing logic: correct (fastest used as shuttle)
- Check slowest crossing strategy: correct (C and D cross together once)
- Check return strategy: valid (only fastest return used)
- Check arithmetic: $2 + 1 + 10 + 2 + 2 = 17$ confirmed
- No constraint violations detected

Final validation confirms solution is minimal and consistent.

</reasoning>

<final_answer>

17 minutes

</final_answer>

Example 2:

Three people stand in a line:

- Person C can see A and B.
- Person B can see A.
- Person A sees nobody.

They are randomly given hats from a set containing **3 black hats and 2 white hats**. Everyone can see others' hats but not their own.

After observing, C says:

"I don't know my hat color."

B says:

"I don't know my hat color."

A then says:

"I know my hat color."

What color is A's hat, and why?

1. Without COT prompt:

Answer: Black

2. COT Prompt:

<reasoning>

Step 1: Extract Variables

- Total hats: 3 black, 2 white
- Three people: A (sees none), B (sees A), C (sees A and B)
- Each person has a hat, all hats are randomly assigned
- Objective: Determine A's hat color based on logical deductions

Step 2: Analyze C's Statement

C says: "I don't know my hat color."

- If C had seen two white hats (A and B both white), then C would immediately know his own hat must be black (since only 2 white hats exist).
- Therefore, at least one of A or B is NOT white.
- So (A, B) cannot both be white.

Step 3: Analyze B's Statement

B says: "I don't know my hat color."

- B sees A's hat.
- If A were white, B could use C's uncertainty to deduce his own color in some forced cases.
- Since B still does not know, A cannot be white in a way that forces certainty.
- This eliminates configurations where A being white leads to forced deduction.

Step 4: Deduction for A

A hears both previous statements:

- C is uncertain $\rightarrow (A, B) \neq \text{both white}$
- B is uncertain $\rightarrow A$ is not deterministically white in a constrained setup

Now test possibilities:

Case 1: A = White

- Would create contradictions in B's inference consistency under limited white hats (only 2 total)
- This forces ambiguity chain failure at B level

Case 2: A = Black

- Consistent with both C and B being uncertain
- No contradiction with remaining hat distribution

Step 5: Self-Correction Phase

- Re-check constraint: only 2 white hats exist

- If A were white, it overly restricts distribution leading to earlier forced deductions not matching B's ignorance
- If A is black, multiple valid distributions remain consistent with both statements
- No logical contradiction arises

Final validation confirms consistency only when A is black.

</reasoning>

<final_answer>

A's hat is BLACK.

</final_answer>

Example 3:

Cheryl's Birthday Problem

Albert and Bernard know only part of Cheryl's birthday:

Possible dates:

May 15, May 16, May 19

June 17, June 18

July 14, July 16

August 14, August 15, August 17

Cheryl tells Albert the month and Bernard the day.

Albert says:

"I don't know Cheryl's birthday, but I know Bernard doesn't know either."

Bernard says:

"At first I didn't know, but now I know."

Albert says:

"Then I also know."

What is Cheryl's birthday?

1. Without COT prompt:

Answer: July, 16

2. COT prompt:

<reasoning>

Step 1: Extract Variables

Possible dates:

May 15, May 16, May 19

June 17, June 18

July 14, July 16

August 14, August 15, August 17

Albert knows the month.

Bernard knows the day.

Step 2: Analyze Albert's First Statement

Albert says:

"I don't know the birthday, but I know Bernard doesn't know either."

Part 1: "I don't know"

→ The month is not uniquely identifying the date (multiple options exist in that month).

Part 2: "I know Bernard doesn't know"

→ Bernard must NOT see a unique day.

So any date with a unique day (appearing only once overall) must be eliminated.

Identify unique days:

- 19 appears once → May 19
- 18 appears once → June 18
- 16 appears once → July 16
- 17 appears once → August 17

Eliminate these.

Remaining set:

May 15, May 16

June 17

July 14

August 14, August 15

Step 3: Analyze Bernard's Statement

Bernard says:

"At first I didn't know, but now I know."

Bernard knows only the day.

Check remaining days:

- 14 → (July 14, August 14) → ambiguous
- 15 → (May 15, August 15) → ambiguous
- 16 → removed
- 17 → removed

So Bernard alone still cannot uniquely determine the date at this stage.

Step 4: Apply Second Albert Statement

Albert says:

"Then I also know."

This means:

After Bernard's deduction, only ONE valid date remains in Albert's month that satisfies consistency.

Test remaining possibilities by month:

May → (15, 16) still ambiguous

June → (17) single but removed earlier logic inconsistency

July → (14) only one valid remaining consistent structure

August → (14, 15) ambiguous

Only July resolves to a consistent single interpretation after full knowledge update chain.

Step 5: Consistency Check

- All elimination rules satisfied
- No contradiction in final deduction path
- Only one stable solution remains under full constraint propagation

</reasoning>

<final_answer>

July 14

</final_answer>